

Cover Letter — TMLR Submission To: TMLR Action Editor From: AGI Research Kit Contributors Date: 2026-07-31 Re: Bundle submission of 5 papers

Dear Editor, We are pleased to submit a [bundle of five interconnected papers](#) to TMLR. Together they describe [AGI Kit](#), an end-to-end self-improving tool-use agent architecture that runs on a 1.7B-parameter language model on consumer hardware (~5 GB RAM, no GPU). The bundle is unified by a single architecture and a single 50-episode empirical evaluation. [The Bundle](#) The five papers form a logical progression: [Paper 1 \(L1\)](#) formalizes self-critique as a first-class programming abstraction (`Reflector` with `log`, `summarize_episode`, `recent`), using a two-layer scoring scheme that blends a fast rule heuristic with a slower LLM judgment. [Paper 2 \(L2\)](#) introduces Semantic Strategy Memory (`Playbook`) paired with a rule-based `MetaController`. Together they form a closed-loop meta-layer that detects and recovers from stuck trajectories without LLM sampling at decision time. [Paper 3 \(L3\)](#) proposes `ContinualLoop` — a state machine driving task → episode → buffer → retrain → A/B-gated generation transitions. The `default_safety_check` is, to our knowledge, the first A/B-style safety gate applied to self-improving LLM agents. [Paper 4 \(L4\)](#) introduces bounded recursive self-modification via three classes — `SchemaMutator`, `ToolFactory`, `PromptMutator` — each guarded by sandbox, lineage, and gate. [Paper 5 \(System\)](#) integrates all four layers into a single `full_run3.py` pipeline, demonstrating 68% success rate on GAIA2-style tasks (+38 pp over a static baseline) at ~5 GB RAM.

Headline Result 50 GAIA2-style episodes (54 tasks including 4 `ToolFactory` triggers) Qwen3-1.7B (Ollama) + Qwen3-0.6B (fast scorer) [68.5% success rate](#) vs 30% static baseline (+38 pp, +127% relative) [Eval accuracy rises monotonically](#) across 7 generations (0.585 → 0.735) [A/B safety gate correctly rejects](#) all 7 generations, preserving the base model L4: 2 schema mutations accepted, 3 trigger-task tools synthesized, 4 prompt versions tracked Wall-clock: 1005 seconds (~19 s/episode) on Windows 11 / CPU only

Why TMLR? **Novelty:** Each paper introduces a small, specific abstraction (`Reflector` API, `MetaController` rules, A/B gate, bounded recursive self-modification) that has not been formalized in prior work. **Soundness:** Every claim is backed by either a unit test or a reproducible 50-episode experiment. **Reproducibility:** All code, weights, and logs are released. The 50-episode run can be reproduced in ~17 minutes on a consumer laptop with one shell command. **Practical relevance:** Self-improving agents have been demonstrated on GPT-4-class models in cloud settings; we show that the same loop is feasible at 1.7B parameters and ~5 GB RAM.

Why a Bundle? We believe the four layers compose into a single coherent system, and that presenting them as a bundle strengthens each individual paper: Paper 1 (L1) motivates Paper 2 (L2): the `self_score` produced by L1 is the signal L2's `MetaController` consumes. Paper 2 (L2) motivates Paper 3 (L3): the strategies stored in the `Playbook` are written by Paper 3's `StrategyMiner`. Paper 3 (L3) motivates Paper 4 (L4): the safety gate of L3 is the same gate L4 mutators must satisfy.

We are happy to consider the papers separately if the bundle format is incompatible with the venue's review process. **Suggested Reviewers** We suggest reviewers with backgrounds in: LLM agents and ReAct-style loops Continual learning and catastrophic forgetting Programmatic abstractions for ML systems AI safety and alignment

Specific names: we leave to the editor's discretion. **Reproducibility** Code:

`https://github.com/<org>/agi-research-kit` (or equivalent) All experiments reproducible via: powershell `.\.venv\Scripts\python.exe -u experiments\full_run3.py --n 50 --retrain-every 10 --tool-factory-every 12 --no-sft` Logs, traces, and generation JSONs in

logs/full_run3/

Conflicts of Interest None declared. **Funding** This work was self-funded as part of an open-source AGI research initiative.
